

TGL-LLM: Integrating Temporal Graph Learning into LLM-Based Temporal Knowledge Graph Forecasting

TGL-LLM (Chang et al., 2025) proposes a novel framework that tightly integrates temporal graph neural networks with large language models to perform forecasting on dynamic temporal knowledge graphs (TKGs). The system aims to improve both temporal pattern modeling and cross-modal alignment between graph embeddings and LLMs.

LLM-Based Temporal Reasoning over Knowledge Graphs (With Examples)

TGL-LLM formulates temporal knowledge graph forecasting (TKGF) as a multi-choice question-answering problem. Given historical TKG snapshots and a query such as:

(Israel, threaten, ?, t+1)

The model is asked to predict the object entity at time $t+1$.

To support reasoning, it first encodes historical events such as:

- (Palestine, assault, Israel, t-2)
- (Israel, assault, Palestine, t-1)
- (Iran, aid, Palestine, t-1)

These are transformed into graph embeddings and aligned with natural language via adapters.

Temporal Graph Embedding with RGCN + GRU (With Examples)

Temporal graphs are modeled as evolving snapshots $G_0, G_1, \dots, G_t$, where each graph G_t contains quadruples of the form (subject, relation, object, timestamp).

Example history:

- t-2: (Iran, aid, Palestine)
- t-1: (Israel, assault, Palestine)
- t: (China, request, Israel)

Graph Encoding:

- Each G_t is encoded using a Relational GCN to capture structural dependencies.

A GRU then aggregates historical embeddings to model temporal evolution:

$$E_{t-2} \rightarrow E_{t-1} \rightarrow E_t \rightarrow \text{GRU} \rightarrow \hat{E}_t$$

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This temporal encoding helps preserve the sequence of events, e.g., that “Iran aided Palestine before Palestine attacked Israel.”

Hybrid Graph Tokenization and Prompting (With Examples)

TGL-LLM introduces a **Hybrid Prompt** that includes:

- Natural language instructions
- Query and candidate facts
- **Graph token sequences** mapped from historical entity/relation embeddings

Example Prompt:

Instruction: Predict the correct object entity given subject, relation, and history.

Query: (Israel, threaten, ?, t+1)

Graph Tokens:

<feature> Iran-aid-Palestine-t-2

<feature> Israel-assault-Palestine-t-1

<feature> China-request-Israel-t

These graph tokens are generated by projecting temporal graph embeddings into token space using *Entity Adapter* and *Relation Adapter* modules.

Two-Stage Training with Influence and Diversity Sampling (With Examples)

To fine-tune the LLM effectively:

- **Stage 1:** Uses high-quality data samples selected via **influence functions** (which estimate how much each sample affects model performance).
Example: Removing (Iran, aid, Palestine, t-1) degrades accuracy → it's influential →

keep it.

- **Stage 2:** Adds a small subset of **diverse samples** to improve generalization across varied temporal and relational patterns.

This two-stage method prevents overfitting to noisy or redundant patterns and ensures broader temporal coverage.

Prediction (With Example)

Given the prompt and graph tokens, TGL-LLM ranks candidate objects:

Candidates: [Syria, Iran, Egypt, Lebanon]

Predicted Answer: Iran

It outputs the answer most consistent with the temporal trends and graph structure.

Evaluation and Results

TGL-LLM was evaluated on the POLECAT dataset across three geopolitical domains (Iran, Israel, Egypt), with timestamps past the LLM's pretraining cutoff to prevent data leakage.

Key findings:

- Outperforms prior Non-LLM (RGCN, ConvTransE) and LLM-based (CoH, KoPA) methods by large margins.
 - Maintains strong performance on **long-tail** (rare) entities due to LLMs' open-world knowledge.
 - Performance improves with longer history (up to ~5 steps), then saturates due to noise.
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Why It Matters for the Survey

TGL-LLM is among the first to **explicitly align dynamic temporal graph embeddings with LLMs** in a forecasting task. Unlike prior works using static graphs or text-only inputs, it:

- Integrates evolving graph structure through GNNs and RNNs

- Embeds this into LLM prompts using a trainable adapter
- Applies curriculum-style training to improve robustness

This makes TGL-LLM highly relevant for the emerging research at the intersection of LLMs and dynamic graph learning.